

Question	Answer	Marks	Guidance
3(a)	thermal energy per unit mass (to change temperature)	B1	Do not allow if context is state change.
	thermal energy per unit change in temperature	B1	Allow 'increase' or 'decrease' for 'change'. <u>Both marking points:</u> Ratio must be clear. Allow 'heat' for 'thermal energy'. Ignore symbols unless defined. Ignore references to units. 'Thermal energy per unit [mass × temperature change]' gains both marks. <u>Special case:</u> Allow 1 mark for use of 'energy' without 'thermal' if the response is otherwise fully correct.
3(b)(i)	$W = p\Delta V$	C1	
	$= 1.01 \times 10^5 \times (1.00 - 0.80) \times 0.0243 = 491 \text{ J}$	A1	Full substitution and answer needed. Allow '1.0 – 0.8' or '0.2' for '1.00 – 0.80'. 80% of 0.0243 may be shown to be calculated separately, and then subtracted from 0.0243. Allow the subtraction either way around. Ignore the sign of the answer. Unit not needed but any unit given must be correct.
3(b)(ii)	volume decreases <u>so</u> (work done on the gas is) positive	B1	Allow 'gas is compressed' for 'volume decreases'. Ignore references to changes in particle separation.

Question	Answer	Marks	Guidance
3(b)(iii)	internal energy is proportional to (thermodynamic) temperature or (thermodynamic) temperature is proportional to volume or internal energy is proportional to volume	B1	Ignore any discussion of work done and transfer of thermal energy. Allow symbols U , V and T without explanation.
	internal energy decreases to 80% of initial value, so increase is (negative and) equal to 0.20×3676 (i.e. -735 J)	B1	For full credit, both B1 marks must come from the same method.
	or		
	ΔU is given by $(3/2)Nk\Delta T$ or U is given by $(3/2)NkT$	(B1)	
	$\Delta U = (3/2) \times 6.00 \times 10^{23} \times 1.38 \times 10^{-23} \times (-0.20 \times 296) = -735 \text{ J}$	(B1)	For full credit, both B1 marks must come from the same method.
3(b)(iv)	$-735 = Q + 491$	C1	Allow C1 for any additive / subtractive combination of (only) 735, 491 and Q .
	$Q = -1226 \text{ J}$	A1	Correct to at least 3 significant figures. AFC. (AFC answer is -1230 J). Answer must be negative for the A1 mark. Answers of $\pm 244 \text{ J}$ or $+1230 \text{ J}$ imply C1.

Question	Answer	Marks	Guidance
3(b)(v)	$\Delta T = 0.20 \times 296$ (= 59 K)	C1	Value of 59 K (to 2 SF) for ΔT implies C1. 296 K – 237 K in the working implies C1. Ignore sign of ΔT . Use of 296 K or 237 K as ΔT is XP in the C1 mark.
	$c = 1226 / (6.00 \times 10^{23} \times 6.64 \times 10^{-27} \times 0.20 \times 296)$ $= 5200 \text{ J kg}^{-1} \text{ K}^{-1}$	A1	Possible ECF from 3(b)(iv) . Correct to at least 2 significant figures. AFC. Ignore sign of answer. <u>Information:</u> The ECF answer from $Q = 244 \text{ J}$ is $1030 \text{ J kg}^{-1} \text{ K}^{-1}$ (but usual ECF rules apply). <u>Important:</u> The answer $1.0 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$ can also be reached by an XP in ΔT , so need to check the working.
3(c)	(same temperature change so) same <u>change</u> in internal energy	B1	Ignore symbol ΔU unless defined.
	(constant volume so) no work done (on/by gas)	B1	Ignore symbol W unless defined.
	less thermal energy supply needed (for same temperature change) so specific heat capacity is less	B1	Allow 'heat' for 'thermal energy'. Allow use of symbols Q and c without explanation (as defined in the question paper).